

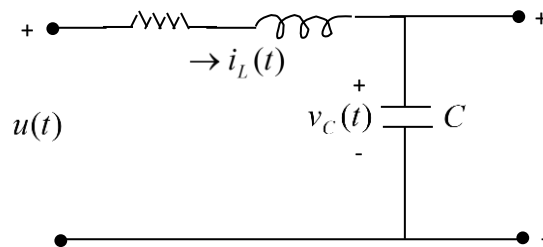
EEEE- 414: Classical Control

Homework Assignment # 3

Due date: by 11:59 PM, Thursday, September 17, 2026

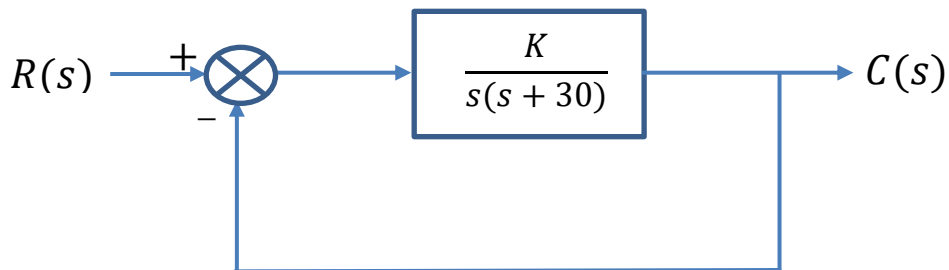
Problem #1

Given the circuit shown below, find the value of the resistor R such that a 20% overshoot in voltage will be seen across the capacitor if $C = 1\mu\text{f}$ and $L=1\text{H}$



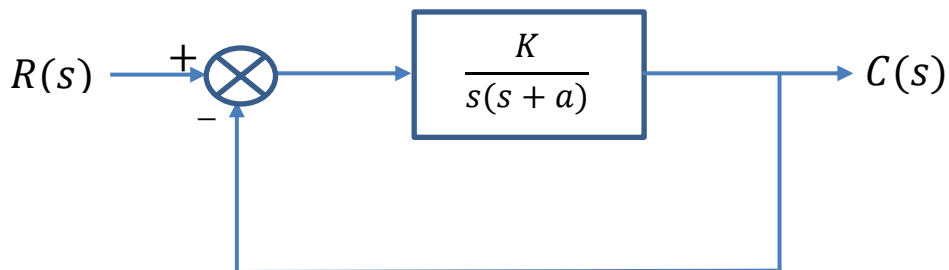
Problem # 2

For the system shown below, find the value of K that yields 10% overshoot for a step input.



Problem # 3

For the system shown below, find K and a to yield a settling time of $0.12[\text{s}]$ and a 20% overshoot for a step input.



Problem # 4

For the system shown below, find the values of K_1 and K_2 to yield a peak time of 1 second and a settling time of 2 seconds for the closed loop system's step response.

